Hospital Data Analysis Using SQL

1) Total Number of Patients

o Write an SQL query to find the total number of patients across all hospitals.

```
SELECT SUM(Patients_Count) AS TOTAL_PATIENTS
FROM hospital_data;
```

	TOTAL_PATIENTS
▶	9347

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2)Average Number of Doctors per Hospital

o Retrieve the average count of doctors available in each hospital.

```
SELECT hospital_name, AVG(Doctors_Count) AS
AVERAGE_COUNT_DOCTOR
FROM hospital_data
GROUP BY Hospital_Name;
```

	hospital_name	AVERAGE_COUNT_DOCTOR
▶	Wellness Clinic	34.0000
	Fortis Care	24.5455
	Heritage Hospital	24.6667
	City Hospital	24.7000
	Global Medicare	28.7273
	Apollo Health	23.3636
	Green Valley Hospital	28.8750
	Healing Touch	25.6429
	Sunrise Medical	28.6250
	Metro Hospital	27.4545

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3)Top 3 Departments with the Highest Number of Patients

o Find the top 3 hospital departments that have the highest number of patients.

```
SELECT Department , SUM(Patients_Count) AS TOTAL_PATIENTS
FROM hospital_data
GROUP BY Department
```

```
ORDER BY TOTAL_PATIENTS desc
LIMIT 3;
```

	Department	TOTAL_PATIENTS
▶	Urology	1766
	Neurology	1229
	ENT	1064

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4. Hospital with the Maximum Medical Expenses
o Identify the hospital that recorded the highest medical expenses.

```
SELECT Hospital_Name, MAX(Medical_Expenses) AS Max_Expenses
FROM hospital_data
GROUP BY Hospital_Name
ORDER BY Max_Expenses DESC
LIMIT 1;
```

	Hospital_Name	Max_Expenses
▶	Healing Touch	49955.41

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5. Daily Average Medical Expenses
o Calculate the average medical expenses per day for each hospital.

```
SELECT Hospital_Name,
ROUND(SUM(Medical_Expenses) /
NULLIF(SUM(DATEDIFF(Discharge_Date,Admission_Date)),0),2) AS
Daily_Expenses
FROM hospital_data
GROUP BY Hospital_Name
ORDER BY Hospital_Name;
```

Hospital_Name	Daily_Expenses
Apollo Health	3187.76
City Hospital	4296.15
Fortis Care	4385.42
Global Medicare	3507.87
Green Valley Hospital	2615.95
Healing Touch	3486.91
Heritage Hospital	2336.24
Metro Hospital	3245.22
Sunrise Medical	4315.76
Wellness Clinic	2830.13

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6. Longest Hospital Stay

o Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

```
SELECT *,
DATEDIFF(Discharge_Date, Admission_Date) AS stay_duration
FROM hospital_data
ORDER BY stay_duration DESC
LIMIT 1;
```

Hospital_Name	Location	Department	Doctors_Count	Patients_Count	Admission_Date	Discharge_Date	Medical_Expenses	stay_duration
Apollo Health	Lucknow	ENT	9	171	2023-04-06	2023-04-21	21666.78	15

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7. Total Patients Treated Per City

o Count the total number of patients treated in each city.

```
SELECT Location, SUM(Patients_Count) AS Total_Patients
FROM hospital_data
GROUP BY Location
ORDER BY Total_Patients Desc;
```

Location	Total_Patients
Jaipur	1505
Ahmedabad	1467
Lucknow	1264
Hyderabad	1261
Bangalore	955
Chennai	840
Kolkata	596
Pune	555
Mumbai	483
Delhi	421

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8. Average Length of Stay Per Department
o Calculate the average number of days patients spend in each department.

```
SELECT Department,
AVG(DATEDIFF(Discharge_Date,Admission_Date)) AS Average_Stay
FROM hospital_data
GROUP BY Department;
```

Department	Average_Stay
Orthopedics	7.1429
ENT	8.0833
Pediatrics	9.1111
Urology	8.7222
Gynecology	7.6667
Oncology	8.1111
General Medicine	7.4286
Dermatology	5.6000
Cardiology	6.8571
Neurology	9.2500

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9. Identify the Department with the Lowest Number of Patients
o Find the department with the least number of patients.

```
SELECT Department, SUM(Patients_Count) AS Total_Patients
FROM hospital_data
```

```
GROUP BY Department
ORDER BY Total_Patients ASC
LIMIT 1;
```

Department	Total_Patients
Cardiology	544

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10. Monthly Medical Expenses Report
Group the data by month and calculate the total medical expenses for each month.

```
SELECT DATE_FORMAT(Admission_Date, '%Y-%m') AS Months,
SUM(Medical_Expenses) AS total_medical_expenses
FROM hospital_data
GROUP BY Months
ORDER BY Months;
```

Months	total_medical_expenses
2023-01	173971.54
2023-02	301722.72
2023-03	199247.42
2023-04	88995.93
2023-05	222986.72
2023-06	165926.36
2023-07	211527.13
2023-08	181039.55
2023-09	341284.23
2023-10	158450.68
2023-11	334370.33
2023-12	337788.45